EdgeGuard-AI

HORIZON EUROPE FSTP

Verifiable Ultra-Low-Latency Edge AI Runtime & Zero-Trust Sensor Verification Engine for Decentralized IoT

Applicant: Abdullah TAY (Dijifabrik) | **Country:** Turkey (Associated Country) | **Requested Grant:** €50,000

1. EXCELLENCE & OBJECTIVES

1.1 Problem Statement

Modern decentralized IoT systems (smart grids, automated manufacturing, telemetry sensor clusters) face critical security vulnerabilities: physical sensor degradation, spoofing injection attacks, and out-of-distribution (OOD) data drift cause AI models to fail silently without raising alerts. Furthermore, transmitting unverified raw telemetry to centralized clouds introduces prohibitive latency, high bandwidth costs, and severe regulatory risks under the EU Data Act and GDPR.

1.2 Innovation: EdgeGuard-AI Architecture

EdgeGuard-AI is an open-source, ultra-low-latency verifiable edge AI runtime designed for resource-constrained edge hardware (industrial gateways, Raspberry Pi, RISC-V). Operating with zero heavy dependencies, EdgeGuard-AI introduces a mathematical zero-trust pre-execution verification engine that analyzes incoming sensor tensors using statistical Z-score, covariance bounds, and rate-of-change physical plausibility checks in under 50 microseconds.

Feature	Standard Edge (ONNX/TFLite)	Heavy Sidecars (K3s)	EdgeGuard-AI (Proposed)
Pre-Inference Tensor Verification	None (Processes all data blindly)	Network-level only	**Mathematical Z-Score & OOD Filter**

Overhead Latency	5 - 20 ms	50 - 200 ms	**< 0.05 ms (48 microseconds)**
Memory Footprint	100 MB - 500 MB	1 GB+	**< 15 MB (Pure Python/C-ABI)**
EU AI Act Audit Attestation	None	Cloud Telemetry Leak	**Local Cryptographic SHA-256 Digest**

2. IMPACT & EU POLICY ALIGNMENT

- **EU AI Act (Article 15 Compliance):** Enforces verifiable robustness, mathematical accuracy, and cryptographic logging on high-risk edge AI systems.
- **EU Data Act & Data Sovereignty:** Enables zero-trust local verification without exfiltrating raw sensor telemetry to external clouds.
- **Green Deal:** Reduces cloud uplink data volume by up to 80%, lowering carbon footprint of industrial IoT deployments.

3. IMPLEMENTATION, WORK PACKAGES & BUDGET

Work Package	Duration	Engineering Hours	Budget	Deliverable
WP1: Core Verification Engine	M1 - M2	130 hrs	€13,000	D1.1: Standalone SensorVerifier module (<50us latency)
WP2: Tensor Quantization Layer	M3	110 hrs	€11,000	D2.1: 8-bit dynamic quantization pipeline (75% RAM reduction)
WP3: Low-Latency Local Broker	M4 - M5	140 hrs	€14,000	D3.1: Priority IPC broker (20,000+ dispatches/sec)

WP4: Open-Source Release & Pilot	M6	120 hrs	€12,000	D4.1: Documentation, Apache-2.0 release, EU data connectors
TOTAL		**500 hrs**	**€50,000**	**Lump-sum 100% Equity-Free Grant**

4. APPLICANT TRACK RECORD & CODE REPOSITORY

Applicant: Abdullah TAY (Head of Digital at Dijifabrik, former Headquarters IT Specialist at HSBC Bank, ITU & Sabancı University Entrepreneurship Alumnus).

Working Prototype & Benchmark: Fully functional codebase with 23 passing unit tests and benchmarked sub-50 microsecond latency is published openly at: https://github.com/abdullahtay/verireach